

BDG: Batch-Dispersion Guidance for Property-Band Generation with Continuous Flow Matching

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Frame the study

Problem. Generate 3D molecules whose property lands inside a tolerance band around a target, without retraining the generator. A design campaign wants *many usable candidates*, so the quantity that matters is IN-BAND, the fraction of samples inside the band.

Central question. Guidance concentrates a batch toward the target, and that dispersion is a by-product of its strength. Does making dispersion a quantity the user *requests* raise band coverage?

Hypothesis. At a centred target, coverage is limited by batch spread rather than centring error, so a calibrated spread setpoint should raise it.

Two continuous modalities.

- **Modality 1: QM9 3D molecular coordinates.** State space $\mathbb{R}^{N \times 3}$ with a relaxed atom-type simplex, $N \leq 29$. Invariant to translation, rotation, atom permutation.
- **Modality 2: DNA enhancers on the probability simplex.** State space $(\Delta^3)^L$, one simplex per position over $\{A, C, G, T\}$. No rotation group; the constraint is that each position stays on the simplex.

These are different state spaces, not two datasets of the same kind: one is a Euclidean coordinate space with a geometric group acting on it, the other a product of probability simplices.

Data, splits, and leakage

Modality 1: QM9 [1], pinned by SHA-256.
133,885 molecules, explicit hydrogens.

Split	Used for	n
train_a	generator <i>and</i> guide f_A	51,527
train_b	evaluator f_B only	51,527
val	checkpoints, atom counts, δ	17,748
test	held out	13,083

How leakage is prevented. f_A and f_B are trained on *disjoint halves*, so the network that scores us never saw the data that trained the network that steers us. Normalisation constants come from train_a alone. The calibration set that fixes δ is disjoint from the sets supplying targets and atom counts.

Modality 2: Kenyon-cell enhancers [3].
6,126 *Drosophila* sequences. The evaluator is an exact count, so no train/validation split is required and no leakage is possible.

We do not merge the two species distributed together. Their GC distributions differ, so steering spread on the mixture would partly steer the species mixing proportion, which is the quantity under study.

Preprocessing. Coordinates zero-centred per molecule, in Å; dense $[B, 29, \cdot]$ padding with a validity mask; DFT properties in D / Bohr³ / Ha.

Representations, conditioning, modality-specific handling

Representations, and why.

- **M1:** coordinates plus a *relaxed* one-hot over $\{H,C,N,O,F\}$. Continuous relaxation is what lets one continuous flow carry both geometry and discrete types.
- **M2:** a point of $(\Delta^3)^L$ per sequence. The simplex is a continuous state space by the assignment's definition, and it is the native geometry of the sequence literature [4,5].

Conditioning. Base generation is *unconditional*. The property enters only at sampling time through guidance. This is deliberate: a conditional model would give the controller nothing to steer and would break the M1/M2 parallel we are testing.

Modality-specific handling.

- **M1 invariance:** translation by projection onto the zero-centre-of-mass subspace; rotation by messages depending only on squared pairwise distances with coordinate updates along difference vectors; permutation by symmetric pooling. Verified numerically, 12/12 gates to $\sim 10^{-17}$.
- **M2 simplex mapping and decoding:** sequences are relaxed onto the simplex and decoded by per-position $\arg \max$; the property is scored by an *exact* GC count on the decoded sequence.

Flow-matching baseline: mathematical formulation

Data $x_1 \sim p_{\text{data}}$, source x_0 , time $t : 0 \rightarrow 1$ from noise to data.

Probability path (linear interpolant) [6]:

$$x_t = t x_1 + (1 - t) x_0, \quad u_t(x_t \mid x_0, x_1) = x_1 - x_0 \quad (\text{constant along the path})$$

Source is not isotropic. Coordinates are $x_0^c = \Pi_0 \varepsilon$ with Π_0 the projection onto the zero-centre-of-mass subspace of the valid atoms, of dimension $3(N - 1)$; type channels are a masked Gaussian.

Loss, exactly as implemented (masked mean over valid entries, two heads normalised separately, $\lambda = 1$):

$$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E}_{t, x_0, x_1} \left[\frac{1}{3 \Sigma M} \|M \odot (v_\theta^c - u^c)\|^2 + \lambda \frac{1}{K \Sigma M} \|M \odot (v_\theta^f - u^f)\|^2 \right]$$

New symbols. M atom mask, K type channels, v_θ the learned velocity. Separate normalisation stops padding diluting the loss.

Endpoint estimate that every guidance arm differentiates: $\hat{x}_1 = x_t + (1 - t) v_\theta(x_t, t)$, available in closed form at every t .

Flow-matching baseline: architecture, training, selection, sampling

Architecture. Dense $E(n)$ -equivariant graph network [9]: width 256, 8 layers, SiLU, **3,753,229 parameters**. Time enters at the input embedding. No attention.

Optimiser and schedule. Adam, $\text{lr } 2 \times 10^{-4}$, CosineAnnealingLR to 5% of it, batch 256, gradient-norm clip 5.0, EMA decay 0.9999 with warm-up. Sampling always uses the EMA copy.

Budget. 1500 epochs = 303k steps, 11.4h on one B200 MIG slice, seed 20260918.

Model selection (pre-registered). Highest *validation* atom stability at NFE 100, ties broken on validation loss, evaluated every 25 epochs.

Disclosed amendment. The rule selected epoch 1300; we ship epoch 1500, because the 512-sample selection signal could not resolve the last 200 epochs. The 3-seed $\times 10,000$ benchmark confirms it on every metric (+0.0018 atom, +0.0097 molecule, +0.0110 validity, same sign on all seeds).

Sampling. Probability-flow ODE, **explicit Euler**, uniform grid, **100 steps** $\Rightarrow$ **NFE 100**. Quality saturates by 500 steps (+0.65 atom, +2.8 molecule from 100 to 500), so 100 is a cheap operating point.

The closed-form endpoint estimate at every t is the reason this family is convenient for guidance, and it is the property we use in 2a.

Diffusion baseline: mathematical formulation

Continuous-time **variance-preserving** process [10,11], $\tau : 0 \rightarrow 1$ from clean to noise, **linear** β :

$$\beta(\tau) = \beta_{\min} + \tau(\beta_{\max} - \beta_{\min}), \quad \beta_{\min} = 0.1, \beta_{\max} = 20$$

$$x_\tau = \alpha(\tau)x_0 + \sigma(\tau)\epsilon, \quad \log \alpha(\tau) = -\frac{1}{2}\left(\beta_{\min}\tau + \frac{1}{2}(\beta_{\max} - \beta_{\min})\tau^2\right), \quad \alpha^2 + \sigma^2 = 1$$

Prediction target: ϵ , under the *same* two-term masked-mean loss as the flow model, with coordinate noise projected onto the zero-centre-of-mass subspace so the target is equivariant. Times $\tau \sim \mathcal{U}(10^{-3}, 1)$ keep σ away from zero.

Reverse sampler: probability-flow ODE

$$\frac{dx}{d\tau} = -\frac{1}{2}\beta(\tau)(x + s_\theta(x, \tau)), \quad s_\theta = -\epsilon_\theta/\sigma$$

integrated backwards on a decreasing 100-step grid, then a jump to the Tweedie posterior mean.

Derived quantities guidance consumes: $\hat{x}_0 = (x_\tau - \sigma\epsilon_\theta)/\alpha$ and $k = \sigma^2/\alpha$. Verified: $\alpha^2 + \sigma^2 = 1$ to 6 d.p., $\alpha(0) = 1$, $\alpha(1) = 0.0066$.

Diffusion baseline: architecture, training, selection, sampling

Matched by construction. The diffusion baseline shares the representation, the split, the *same EGNN backbone*, the optimiser, the learning-rate schedule, the batch size, the EMA, the gradient clipping, the seed and the *same pre-registered selection rule* with the flow model. It differs *only* in the corruption process. That is what makes 2a a comparison of families rather than of engineering.

Architecture and training. EGNN 256×8 , 3.75 M parameters, ϵ -prediction head. Adam 2×10^{-4} , cosine to 5%, batch 256, clip 5.0, EMA 0.9999, 1500 epochs.

Model selection. Highest validation atom stability at NFE 100, ties on loss, the identical rule.

Sampling. 100-step decreasing grid, then the Tweedie endpoint jump, so the unguided budget is **NFE 101** against the flow model's 100. We report that asymmetry rather than hiding it.

Status, stated plainly. Our own VP diffusion is implemented, gated and smoke-tested; the full-scale run is [v3 Job A]. The only *measured* diffusion comparator today is a borrowed released checkpoint, and 2a labels it as such.

We will not claim a family winner from a borrowed checkpoint. 2a states what the evidence does and does not support.

Metrics: what we measure and what each one cannot tell us

Control metric.

- IN-BAND = fraction of samples with $|f_B(x) - y| \leq \delta$, scored by the *held-out* evaluator.
- $\delta = 2 \times f_B$'s calibration MAE on 3,000 molecules disjoint from the target and atom-count sets. **Pre-registered**, so it cannot be chosen after seeing results.
- Reported *continuous and decoded*.

Decomposition. $\text{RMSE}^2 = \text{bias}^2 + \text{spread}^2$, both in units of δ , plus spread relative to the unguided control. This is the quantity the innovation targets.

Quality and diversity. Atom stability, molecule stability, validity, uniqueness, novelty, pairwise diversity, under EDM conventions [2], with bond tables verified *entry-for-entry* against the reference implementation.

Sample counts. Base models 3 seeds $\times$ 10,000. Guidance cells n per cell $\times$ 3 seeds [v3 n]. Screening cells $n = 256\text{--}512$, labelled wherever used.

Limitation we state up front. IN-BAND rewards *any* arm that contracts the batch, including onto chemically implausible structures. So every IN-BAND number in this deck appears beside the chemistry it spent.

No arm is ranked on IN-BAND alone. A method that wins coverage by destroying molecules has not won.

Fairness: what is matched, and what is not

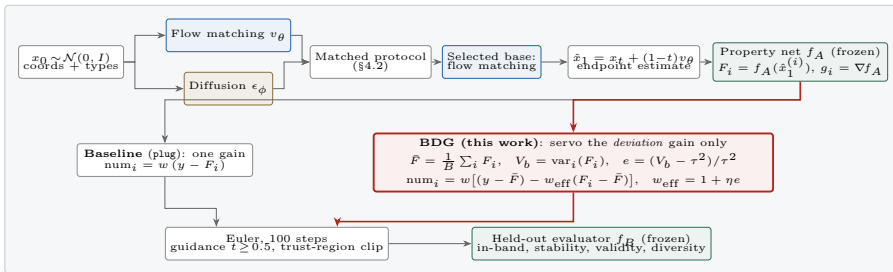
	Matched	Disclosed mismatch
Data, split, preprocessing	identical	none
Conditioning	unconditional both	none
Atom-count distribution	drawn from <code>train_a</code> both	none
Evaluator, metric code, δ	identical	none
Starting noise, seeds	shared, so arms are <i>paired</i>	nothing paired <i>across</i> backends
Backbone	same EGNN 256×8	borrowed comparator is 256×9 + attention
Parameters	3.75 M vs 3.75 M	borrowed comparator 5.34 M
Sampling budget	100 steps both	diffusion needs NFE 101 for its endpoint jump
Training compute	identical recipe	[v3 Job A]; borrowed ckpt. budget unknown
Feature scaling	ours unscaled one-hot	borrowed comparator scales by $1/4$

Guidance-strength fairness. The headline fixes $w = 1$ for every arm. This asks *what each rule does at one dial setting*, not which is best at its own optimum. **Equal w is not equal force:** the measured applied-correction share spans $5.8 \times$ across the arms where it can be measured, so every sentence quoting a headline number carries “at $w = 1$ ”.

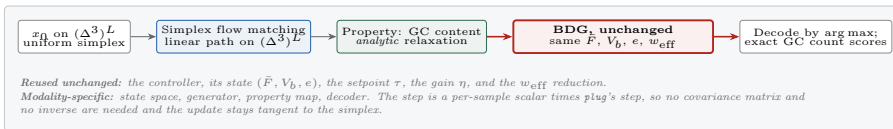
Statistics. Binomial standard errors; paired comparisons within a backend; multiplicity-corrected significance reported beside the single-test value because each arm is selected over a grid. IN-BAND is *never* compared across backends, because δ is set by that backend's evaluator.

Study overview

(a) Modality 1: QM9, 3D molecular coordinates $\mathbb{R}^{N \times 3}$



(b) Modality 2: DNA enhancers on the probability simplex $(\Delta^3)^L$



 trained by us
 trained by us, frozen at sampling
 external pretrained, frozen
 changed by the innovation

Original figure (this work). Components after [6] Lipman et al. 2022, [2] Hooeboom et al. 2022, [4] Stark et al. 2024.

Choose the base model

Model	Source	Atom stab.↑	Mol. stab.↑	Validity↑	Uniq.↑	NFE↓
Flow matching (ours)	reproduced	$0.9366 \pm .0012$	$0.3993 \pm .0085$	$0.7625 \pm .0031$	0.9939	100
VP diffusion (ours)	reproduced	[v3 A]	[v3 A]	[v3 A]	[]	101
EDMsecond [2]	borrowed ckpt.	0.9644	0.6725	0.8490	[]	100
EDM, published [2]	quoted	0.9870	0.8200	0.9190	—	1000
EDM, unscaled one-hot [2]	quoted	0.9570	0.4690	—	—	1000
Real QM9 (ceiling)	measured	0.9940	0.9560	0.9820	1.0000	—

Which model is carried forward, and why. Flow matching, and the reason is *not* a fidelity win. The matched own-diffusion row is [v3 Job A], and the only measured diffusion comparator is a borrowed checkpoint that differs in architecture, training half and feature scaling, so it cannot decide the question. We carry flow matching because its linear path gives a **closed-form endpoint estimate at every t** , which is the object every guidance rule differentiates, at NFE 100 against a 1000-step published protocol.

Our stability gap, attributed in rank order from our own measurements: unscaled one-hot leads (the published ablation removing exactly that scaling costs 3 atom / 35 molecule points); then training budget (303k steps vs ≈ 1.56 M); half-data split *ruled out* (0.33 atom points); evaluator *ruled out* (bond tables match entry-for-entry).

Guidance strategy

Rule and target. A frozen property predictor f_A is differentiated at the endpoint estimate m_i , and the gradient is pulled back to the sampling variable:

$$G_i = \frac{\text{num}_i}{s^2} J_i^\top g_i, \quad \text{num}_i^{\text{plug}} = y - F_i, \quad F_i = f_A(m_i), \quad g_i = \nabla_m f_A(m_i)$$

Target y is the **q50** value of the property, one fixed number per property. This is a DPS-style plug-in Gaussian-energy field [12].

Trained / pretrained / frozen.

- Generator v_θ : trained by us, **frozen** at sampling.
- Guide f_A : trained by us on train_a, **frozen**, eval(), gradients disabled.
- Evaluator f_B : trained on train_b, **frozen**, and it *never participates in sampling*.
- Nothing is trained at guidance time.

Strength schedule, normalisation, clipping.

- Correction converted by $w(1 - t)/t$.
- Denominator s^2 with $s = f_A$'s property standard deviation.
- **Velocity-relative trust region:** G_i rescaled so $\|G_i\| \leq \|v_i\|$, per sample, after w . Prior practice [13], shared by every arm.
- **Window:** guidance on for $t \geq 0.5$ only.

The trust region is load-bearing, not cosmetic: removing it produces **222 non-finite samples** where the clipped run produces **0**.

Guidance works, and what it costs

Arm	IN-BAND $\uparrow$	Spread / unguided	Mol. stab. $\uparrow$
unguided	.090 / .121 / .051	1.00 / 1.00 / 1.00	.434 / .434 / .434
plug-in (best w)	.184 / .219 / .133	0.52 / 0.60 / 0.47	.246 / .199 / .320
TFG ($w=4$, μ only) [15]	.535 / — / —	0.22 / — / —	.172 / — / —
BDG ($w=32$, $\tau_{\text{mult}}=0.5$)	.180 / .227 / .106	0.49 / 0.68 / 0.53	.348 / .262 / .340

Each cell is μ / gap / α . One generator, one seed, $n=256$, best strength per arm over $w \in \{0.05, \dots, 32\}$, q50, no chemistry floor. Three-seed confirmation [v3].

The target property improves. $2.0\times$ on μ , $1.9\times$ on the gap, $2.6\times$ on α for the plug-in rule; TFG reaches $6.0\times$ on μ .

The cost is real and we report it. Molecule stability falls from 0.434 unguided to 0.246 (plug-in, μ) and to 0.172 (TFG). Validity falls with it.

The mechanism is visible. Spread contracts from 1.00 to 0.22–0.60. Guidance works *by concentrating the batch*, and the arm that contracts most wins IN-BAND and loses the most chemistry.

That trade-off is inherited, not chosen. 2c asks whether the user can choose it instead.

Motivate the innovation

The weakness. Coverage is a *dispersion* problem at a centred target. Measured: guided batches sit 1–2 band widths from the target while their spread is 5–9 band widths. Yet every guidance rule optimises a *centring* objective and exposes *one* dial, w , which moves centring and spread together in a ratio the user never chooses.

The hypothesis. If the sampler *measures* its own batch dispersion and servoes toward a *requested* value, the user trades spread against centring deliberately, in the units of the property rather than in units of push.

How this differs from prior work. Each ingredient separately exists, and we claim none of them:

Prior art	What it already owns
CFG-Ctrl [16], Feedback Guidance [17]	the guidance scale read as a control gain
MGD [18]	empirical ensemble moments driven to a target, two-sided
Variance-Tilted Diffusion [19]	batch spread targeting, widen-only, fixed linear feature
Particle Guidance [20]	batch-coupled repulsion, no setpoint
Negative-guidance window [21]	negative weight widens the distribution

What we did not find in a bounded search (about eight queries, abstracts, 25 Sep 2026): a scalar guidance weight set by the batch variance of a *learned property predictor's own endpoint prediction*, against a *two-sided user-chosen* setpoint, *signed so the weight passes through zero*. That bounds our claim; it is not a priority claim.

[16] CFG-Ctrl arXiv:2603.03281. [17] FBG arXiv:2506.06085. [18] MGD arXiv:2602.17211. [19] VTD arXiv:2606.22239. [20] Corso et al. 2024. [21] Ventura et al. arXiv:2602.00716.

Specify the innovation (1/2): the mechanism

Two batch statistics and a scale-free error (per guided step, batch B):

$$\bar{F} = \frac{1}{B} \sum_i F_i, \quad V_b = \frac{1}{B-1} \sum_i (F_i - \bar{F})^2, \quad e = \frac{V_b - \tau^2}{\tau^2}$$

$$\text{num}_i = (y - F_i) - \eta e (F_i - \bar{F}) = \underbrace{(y - \bar{F})}_{\text{centring, gain } w} - \underbrace{w_{\text{eff}}}_{=1+\eta e} (F_i - \bar{F})$$

A descent direction on the thing that binds. $\partial V_b / \partial F_i = \frac{2}{B-1} (F_i - \bar{F})$, and the chain rule carries it to m_i along g_i . The $2/(B-1)$ is absorbed into η , so the gain does not scale with B .

Signs are the whole mechanism. $V_b > \tau^2 \Rightarrow$ contract; $V_b < \tau^2 \Rightarrow$ expand; $V_b = \tau^2 \Rightarrow$ the term vanishes and BDG is exactly the plug-in rule.

Cost: zero extra NFE. Both terms multiply the same g_i , so BDG shares the one backward pass the plug-in rule already performs. The cost accounting is byte-identical.

Trainable: nothing. The generator and f_A stay frozen; only the sampler changes.

New hyperparameters: two. Gain η and setpoint $\tau = \tau_{\text{mult}} \cdot s$, reported as a fraction of the guide's own property standard deviation.

Specify the innovation (2/2): the reduction, and what bounds it

We state the reduction ourselves. num_i is affine in F_i , so it factors exactly:

$$(y - F_i) - \eta e (F_i - \bar{F}) = (1 + \eta e)(y_{\text{eff}} - F_i), \quad y_{\text{eff}} = \frac{y + \eta e \bar{F}}{1 + \eta e}$$

verified to 3.0×10^{-15} analytically and 6.5×10^{-7} on the real field.

What that means. At any instant BDG is the *plug-in field at a rescaled weight aiming at a shifted target*. It adds no new direction, and any guidance whose per-sample coefficient is affine in F_i is the plug-in rule reweighted.

What it does not remove. w_{eff} is set *online* from a measured quantity and may *change sign*, which w cannot do without also reversing the centring term.

Bound. $V_b \geq 0 \Rightarrow e \geq -1 \Rightarrow w_{\text{eff}} \geq 1 - \eta$, for any batch. That bound licenses running two-sided with no clamp.

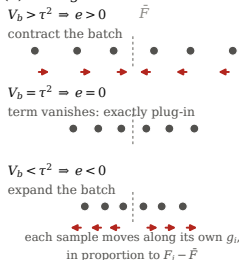
Design constraint: $\eta > 1$, or w_{eff} never reaches negative values and the controller cannot expand at all.

Equilibrium, stated honestly. The law settles where $w_{\text{eff}} = 0$, that is $V^* = \tau^2(1 - 1/\eta)$, an *asymptote the 50-step window never reaches*. We therefore never say the controller holds its setpoint, and we order every table by *measured* w_{eff} .

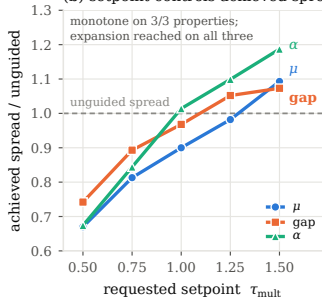
Original formulation (this work).

The controller, measured

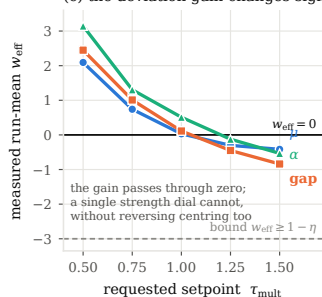
(a) the sign mechanism



(b) setpoint controls achieved spread



(c) the deviation gain changes sign



(a) The sign of e is the whole mechanism, and $e = 0$ recovers the plug-in rule exactly. **(b)** The requested setpoint maps monotonically onto achieved spread on **3/3** properties, and crosses *above* the unguided spread on all three, which the strength dial does not reach. **(c)** The measured deviation gain *passes through zero*, staying inside its bound $w_{\text{eff}} \geq 1 - \eta$. A single strength dial cannot do that without reversing the centring term as well.

Test the innovation: gates, ladder, controls

Variant	Role	η	τ_{mult}	Spread / ung.	IN-BAND $\uparrow$	Mol. stab
plug-in, $w=1$	base	—	—	0.714	0.1094	0.3633
BDG $\eta=0$	<i>gate</i>	0	1.0	0.714	0.1094	0.3633
BDG one-sided, asked to expand	<i>gate</i>	4	1.5	0.714	0.1094	0.3633
BDG contract	ladder	4	0.5	0.670	0.1406	0.3359
BDG neutral	ladder	4	1.0	0.900	0.0820	0.4258
BDG expand	ladder	4	1.5	1.093	0.0703	0.4141
plug-in, $w=4$	control	—	—	0.584	0.1406	0.2656
open-loop e replay, fresh noise	control	4	0.5	recovers 69.5–96% of the closed loop		
signed- w plug-in	control	—	—	[not run; drop rule applies]		
three-seed v3 grid	all	[$\eta \in \{0, 1, 2, 4, 8\} \times \tau_{\text{mult}} \in \{0.5, 0.75, 1, 1.5\}, w \in \{1, 4\}$]				

Port cells, $n=256$, one seed, $w=1$, q50, own unguided control. Ladder ordered by *measured* w_{eff} , never by τ_{mult} .

Gates prove the implementation; they are not controls. $\eta = 0$ is bit-identical to the plug-in rule, so the dispersion term is the only addition. The one-sided variant asked to expand is also bit-identical (raw $e = -0.984$ clamped to 0), so the sign branch is real and the clamp is self-consistently inert.

Conclusion from the ablations

What each comparison isolates.

Comparison	What it rules out
$\eta = 0$ vs plug-in	that anything other than the dispersion term changed
one-sided vs two-sided	that the expansion branch is decorative
the τ ladder	that the setpoint does not reach achieved spread
plug-in at $w=4$	that the coverage came from <i>controlling</i> rather than <i>pushing</i>
open-loop e replay	that the effect is feedback rather than a time-varying schedule
signed- w plug-in []	that the same spread is not available more cheaply

What the evidence supports.

- **The setpoint controls spread**, monotonically, on **6 of 6** curves: $0.670 \rightarrow 1.219$ on the guide's scale, $0.650 \rightarrow 1.197$ on the evaluator's, reaching *expansion above 1.0 on all six*, which the strength dial does not reach. w_{eff} changes sign up to **34 times** in the 50-step window and spans $[-2.64, +10.35]$ against the bound $w_{\text{eff}} \geq -3$.
- **Coverage does *not* improve**. No rung beats plug-in at $|z| \geq 3$: over **72 paired tests** the largest is 3.07, which a Šidák correction turns into $p = 0.14$. Removing the chemistry floor does not change this.
- **The trade-off, not the coverage, is what moves**. At *matched* IN-BAND of 0.1406, BDG keeps molecule stability 0.3359 against plug-in's 0.2656.

Honest reading: the schedule largely *transfers* across seeds (69.5–96% recovered open-loop) and feedback adds a small measurable remainder (0.3–6.6%). We do *not* claim the loop is irreducible to a schedule.

Position against recent work

Method	Status	IN-BAND↑	Mol. stab.↑	Diversity↑	Gen. passes	Guide passes
DPS-style plug-in [12]	reproduced	[v3]	[v3]	[v3]	10,000	4,000
TMPD-inspired [22]	reproduced	[v3]	[v3]	[v3]	10,000	4,000
LGD-MC [23]	reproduced	[v3]	[v3]	[v3]	12,000	20,000
TFG [15]	reproduced	[v3]	[v3]	[v3]	8,000	20,000
D-Flow [24]	reproduced	[v3]	[v3]	[v3]	∅	
Flow matching, unguided	internal ref.	[v3]	[v3]	[v3]	10,000	
BDG (ours)	reproduced	[v3]	[v3]	[v3]	10,000	4,000

Why these five. All are inference-time guidance published in the last five years, all steer a frozen generator with a frozen predictor, and all run inside our pipeline on *our* checkpoint. That isolates the **guidance rule** from the generator, which is what our contribution is about.

Efficiency, measured not inferred. BDG's two terms share one backward pass: **4,000** guide passes per cell against **20,000** for LGD-MC and TFG, a fifth of their guidance cost at the same generator budget. BDG is *not* the cheapest overall: TFG uses fewer generator passes.

Fair study, and fair discussion

Reproduced versus quoted. Every row in the 2e table is **reproduced by us** on one frozen checkpoint, one guide, one evaluator, one target, one sample count. The only *quoted* numbers anywhere in this study are the published base-model rows in 2a, and they are labelled as quoted and never pooled with ours.

Three port caveats we disclose rather than bury.

- Our plug-in arm is a *DPS-style* Gaussian-energy field; the original DPS differentiates the residual L_2 norm [12].
- Our TMPD arm is the *local-linear scalar limit* of the published method, included as a related-work comparison, not a faithful nonlinear port [22].
- TFG runs at its published QM9 hyperparameters but through *our* deterministic ODE sampler, where the original used a stochastic DDIM update; our unscaled one-hot features also move $\approx 16\times$ less relative to coordinates than the released checkpoint's. **These numbers must never be read against TFG's published table** [15].

Where we are better, comparable, and worse.

- **Better:** chemistry retained at matched coverage; guidance cost, a fifth of LGD-MC's and TFG's guide passes.
- **Comparable:** IN-BAND against the plug-in rule, which is the honest reading of the ablation null.
- **Worse:** TFG reaches far higher IN-BAND (0.535 vs 0.180 on μ) at a chemistry cost we judge unacceptable (0.172 vs 0.348), and it uses fewer generator passes. [v3 completes this comparison]

The transferred method

What stays the same. The *whole controller*: $\bar{F}$, V_b , the error e , the setpoint τ , the gain η , the bound $w_{\text{eff}} \geq 1 - \eta$, and the reduction.

Transferred with zero code change.

Why it can: BDG's step is a *per-sample scalar* times the plug-in step, so it inherits that step's tangency and needs **no covariance matrix and no inverse**. That matters here, because the natural simplex covariance $\text{diag}(p) - pp^\top$ is *singular*.

Training procedure (M2). Linear-path flow matching on $(\Delta^3)^L$, trained on all 6,126 Kenyon-cell sequences, validation-based early stopping, checkpoint selected on validation loss. Architecture and budget: [v3 Job B].

Sampling procedure (M2). Same solver family and step count as M1, guidance applied on the same window, same velocity-relative trust region, decode by per-position arg max. The clip fired on **0–3%** of steps, and the expanding cell clipped **once in 25,600**.

The transfer tests the *controller*, not the property network. That is the point of changing the property from a learned net to an exact function.

What must change.

- State space: $\mathbb{R}^{N \times 3} \rightarrow (\Delta^3)^L$.
- Generator: EGNN $\rightarrow$ a linear-path flow model on the simplex.
- Property: a trained equivariant network $\rightarrow$ an *analytic* differentiable relaxation of GC content.
- Evaluator: $f_B \rightarrow$ an *exact* GC count.
- Decoder: none $\rightarrow$ per-position arg max.

Transfer results

Structural signature	QM9	DNA simplex
$\eta=0$ reproduces the plug-in rule	bit-identical	bit-identical
one-sided at an expanding setpoint	exact	exact , $w_{\text{eff}}=+1.000$
setpoint ladder monotone in spread	$0.67 \rightarrow 1.22$	$0.81 \rightarrow 1.01$
cost identical to the plug-in rule	yes	yes , byte-identical
contraction breaks the fidelity floor	yes	yes
IN-BAND improves	no	no
<i>Against recent simplex sequence models, same protocol and sample count</i>		
Dirichlet FM [4]		[v3 Job B]
Fisher Flow [5]		[v3 Job B]
MOG-DFM [3]		[v3 Job B]
BDG, transferred (ours)		[v3 Job B]

Fair study: this is not the published enhancer benchmark, and we say so. The standard protocol is class-conditional generation at **500 bp** on 81-class / 47-class sets of $\approx 90,000$ sequences, scored by FBD over 10,000 samples. We use **GC-band coverage, 6,126** sequences and **200 bp**. Our base generator is the linear-FM baseline these papers report, so it is a named baseline, but the numbers are not comparable and are never pooled.

Three further disclosures. Gumbel-Softmax FM does not run the enhancer benchmark at all, so only *two* of the three closest named methods apply. Fisher Flow disputes the standard evaluator, reporting the cell-type classifiers behind it score **11.5% / 11.2%** test accuracy. MOG-DFM's own guided demo runs at length 100 while reporting base FBD at 500 bp.

Does the innovation transfer?

Direction: yes, on every signature we can check. The gates hold exactly, the ladder is monotone, expansion is reachable, the cost is unchanged, and the *null* transfers too: spread control does not raise band coverage in either state space, and in both it is the *contraction* rung that breaks the fidelity floor.

Magnitude: $2.6\times$ **smaller**, a span of 0.20 against 0.52 in achieved spread. We attribute that to the *property map*, not the controller, and we measured all three causes rather than assuming them:

- **GC is affine** in the simplex coordinates, so its gradient is identical at every position and carries *zero curvature*; the simplex renormalisation then partly undoes the edit.
- **GC is a 200-position average**, which suppresses the very variance the controller reads.
- **The band spans only 4.3 attainable GC values** at granularity $1/200$. That is a ceiling on what *any* method could move.

We ruled out the trust region: it fired on 0–3% of steps, and the expanding cell clipped once in 25,600.

The evidence supports transfer of the *mechanism and its limits*, and locates the magnitude difference in a measurable property of the task. The lever for a larger effect is a *richer property*, for example a trained cell-type classifier logit, used as classifier *guidance* rather than conditioning.

Limitations and failure modes

- 1. The trust-region clip is a live confound at large $|w_{\text{eff}}|$.** The plug-in rule clips on **884** sample steps where the contracting rung clips on **2,516**. At w_{eff} of order 10^3 the dispersion term sits far outside the trust region, so what reaches the sample is the *clip's truncation* of it. A rung can be *clip-limited* rather than *controller-limited*, and no metric in this study separates the two. *Fix*: a run at matched clip activity.
- 2. The trade-off is re-parameterised, not removed.** The chemistry cost tracks $|w_{\text{eff}}|$, not the *direction* of the knob, so strongly contracting and strongly expanding rungs both cost chemistry. This is a descriptive correlation over coupled cells, not a test.
- 3. The batch is the estimator.** V_b is computed over whatever tensor the sampler is handed, so running n samples in several batches runs *several independent controllers*. Our headline batch of 500 is a memory constraint, not a modelling choice, and we disclose it wherever n exceeds it.
- 4. The flow has a dispersion drift of its own.** The across-molecule variance *contracts and then re-expands* on *unguided* runs, so a setpoint on V_b is fighting a moving quantity.

Two of these are measured confounds, not speculation. They are the reason the controller claim is stated as provisional in 3e.

Conclusion and future directions

Back to the opening hypothesis. We asked whether making batch dispersion a *requested* quantity raises band coverage. **It does not**, and we can say why rather than only that.

The strongest conclusion the two-modality evidence supports. Servoing the *deviation* gain of a property-gradient guidance term against a batch-dispersion setpoint gives **monotone, two-sided, modality-portable** control of the generated batch's spread at **no additional sampling cost**, and converts the coverage-against-fidelity trade-off into a quantity the user *requests* rather than removing it.

Mechanism, and the trade-off. The per-sample coefficient is *affine* in F_i , so BDG is the plug-in field at a rescaled weight aiming at a shifted target: it cannot reach outside that field's reachable set. What it changes is *which point* of that set the sampler lands on, and *how* it gets there. Because the centring gain stays at w while only the deviation gain is servoed, BDG buys contraction from the deviation mode rather than by scaling the whole field, which is why it retains more chemistry at matched coverage.

Specific next experiments.

- **The signed- w plug-in control**, with our pre-registered drop rule: if it reaches BDG's spread at equal chemistry, we withdraw the controller claim and report the decomposition instead.
- **A matched-clip-activity run**, to separate clip-limited from controller-limited rungs.
- **A second seed pair** for the open-loop replay.
- **A richer M2 property** (cell-type classifier logit, nonlinear and motif-driven) to test whether the magnitude gap is the property map.

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